

CD Burning Workstation

Chain K — Hardware & Systems Foundations · Full written lesson

The old machine from Linux-On-Old-Hardware is sitting there with a working Linux install and an optical drive nobody uses anymore — putting it to work as a dedicated burn station is a genuinely satisfying project.

Lesson 1 — Choosing Blank Media, and Confirming Your Drive Can Actually Burn

Before downloading a single track, two things need confirming: what kind of disc actually fits the job, and whether the drive in this old machine can write one at all.

- CD-R is write-once, using a photosensitive organic dye. CD-RW is rewritable, using a phase-change metal-alloy film — but many car stereos and older standalone players genuinely cannot read CD-RW discs at all.
- Capacity comes in two standard sizes: 650MB/74min or 700MB/80min.
- Dye quality matters for longevity: archival-quality phthalocyanine-dye CD-Rs can exceed 100 years under good conditions; cheap, non-archival discs can degrade in as little as 5-10 years.
- Many optical drives — especially in a machine repurposed as "just a Linux box" — are read-only. Confirm real burn capability with `wodim --devices` before buying any media at all.

The trap: buying the cheapest unbranded CD-R spindle for something meant to last, and assuming every optical drive can burn.

Worked example: running `wodim --devices` on a drive that turns out to be read-only saves buying a stack of blank discs for hardware that can never write them.

Lesson 2 — Downloading With yt-dlp, and Converting to the Exact Format Audio CD Requires

This is the step where "any audio file" stops being good enough — Audio CD has one fixed, exact target format.

- `yt-dlp` is an actively maintained downloader; `-x --audio-format <fmt>` extracts just the audio.
- Audio CD (Red Book / CD-DA) requires exactly 44.1kHz sample rate, 16-bit depth, and 2-channel (stereo) PCM audio.
- `ffmpeg` converts to that exact spec: `-ar 44100 -sample_fmt s16`, writing a WAV file.
- Verify with `ffprobe` before touching a blank disc.
- Package compatibility: `yt-dlp`, `ffmpeg`, and `wodim/genisoimage` (the `cdrkit` package) install via `apt` on Debian/Xubuntu or `apk` on Alpine — same tools, different package manager.

The trap: burning straight from a downloaded MP3/AAC without converting to the exact PCM format Audio CD needs. Separately: GUI burners (Brasero/K3B) pull in real GNOME/KDE library weight — a conflict with a

lightweight-distro machine. CLI tools are the actual recommended path.

Worked example: a downloaded file at 48000Hz/24-bit/mono fails all three Audio CD requirements at once; running it through `ffmpeg -ar 44100 -sample_fmt s16` makes it fully compliant.

Lesson 3 — Audio CD vs. Data CD, and Verifying a Burn Before Trusting It

Two genuinely different disc types exist for two different goals — and a burn that "finished" isn't automatically a burn you can trust.

- Audio CD (via `wodim / cdrecord -audio`) plays in any standalone CD player.
- Data CD (via `genisoimage / mkisofs`) stores files as a regular filesystem — computer/MP3-player playback only, but more content per disc.
- TAO introduces a real ~2 second gap between tracks; DAO burns the whole disc continuously with no gap.
- A burn that completes with no reported error is not the same as a burn that's actually readable end to end.

The trap: not verifying the burn. A disc that appears to finish can still contain real read errors.

Worked example: the same track list burned in TAO mode has audible gaps between songs; the identical files burned in DAO mode play back gap-free.

Lesson 4 — Scripting the Whole Pipeline, and Where Personal Archival Ends

This is the natural place to actually use the shell fluency from Shell-And-Small-Software instead of just drilling it.

- Wrap download → convert → burn in one script with real argument validation and clear exit codes.
- `set -euo pipefail` belongs here too: a failed download shouldn't silently let the script burn whatever partial file resulted.
- Ethics, precisely: this project is for personal-use archival of your own material or content you have the rights to — not redistribution or piracy.

The trap: downloading and redistributing copyrighted content. The personal-use line is a real, meaningful one.

Worked example: a script that validates the URL list and disc size before downloading anything, exiting with a clear nonzero code if the list is empty or malformed.

Reflection

What's actually gained by burning a physical disc in 2026 versus keeping a folder of files or just streaming?
Where's the line, for you, between personal archival and something you shouldn't be downloading at all?